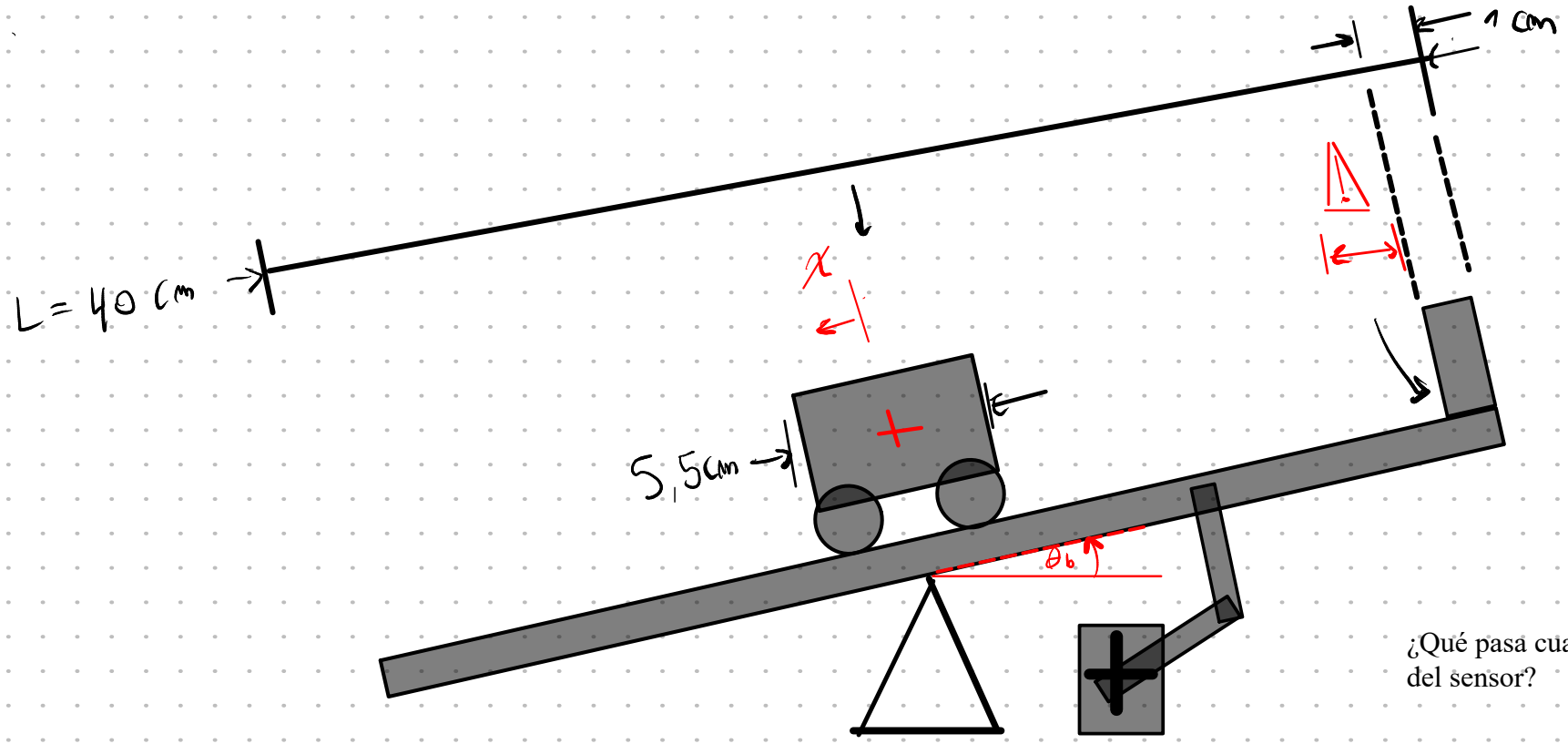


¿Qué pasa cuando el carrito está a menos de 2cm del sensor?

Comando: $u = \theta_s \rightarrow$ ángulo del servo (en °) $\quad \theta_s|_{eq} = 0°$

Salida: $x \rightarrow$ posición del carro $\quad x_{eq} = 0$ cm

θ_b: positivo en sentido A.H.

$$\boxed{x = \hat{L} + \frac{5,5}{2}}$$

Incluye el rango muerto del ultrasónico

L: lectura del u. sónico

$\hat{L}$: " mapeada al sist. de ref.

x: posición del centro del carrito

Valores límite

$\theta_s \rightarrow$ medir

$$x_{min} < x < x_{max}$$

$$x_{max} = 20\,cm - \frac{5{,}5\,cm}{2} = 17{,}25\,cm$$

$$x_{min} = -20\,cm + 1\,cm + \frac{5{,}5\,cm}{2} = -16{,}25\,cm$$ (+2 cm del tango muerto del sensor)

$\boxed{\theta_b/\theta_s}$ ✓ $\boxed{x/\theta_s = x/\theta_b \cdot \theta_b/\theta_s}$ Valores límite → hacer en el laboratorio

$$\theta_b/\theta_s = \frac{K}{(s-p_1)(s-p_2)}$$

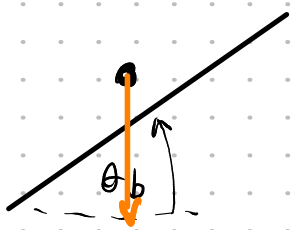
$K = 138{,}3035$

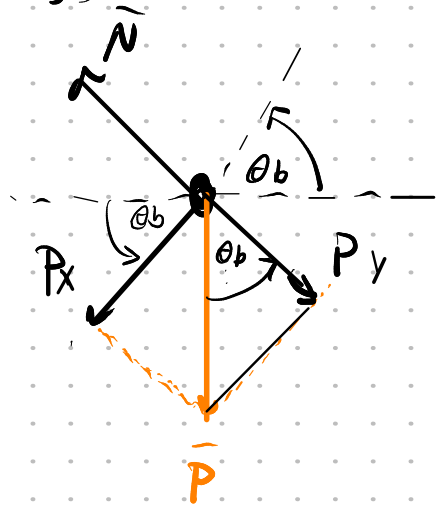
$p_1 = -13{,}2515$ rad/s $\longleftrightarrow -2{,}109$ Hz

$p_2 = -32{,}6059$ rad/s $\longrightarrow -5{,}1894$ Hz

Carrito → masa puntual. No hay F_{roz} ni F_{vis}

m → masa en kg (la medimos)





$P = m \cdot g$

$P_x = m \cdot g \cdot \operatorname{sen} \theta_b$

$P_y = m \cdot g \cdot \cos \theta_b$

$$\int_0^{+\infty} 1 \cdot e^{-s \cdot t}\, dt = \frac{-1}{s} e^{-st}\Big|_0^{\infty} = \frac{-1}{s}(0-1) = 1/s$$

Primer modelo

$N: \cancel{m} \cdot \ddot{x} = \cancel{m} \cdot g \cdot \operatorname{sen} \theta_b$

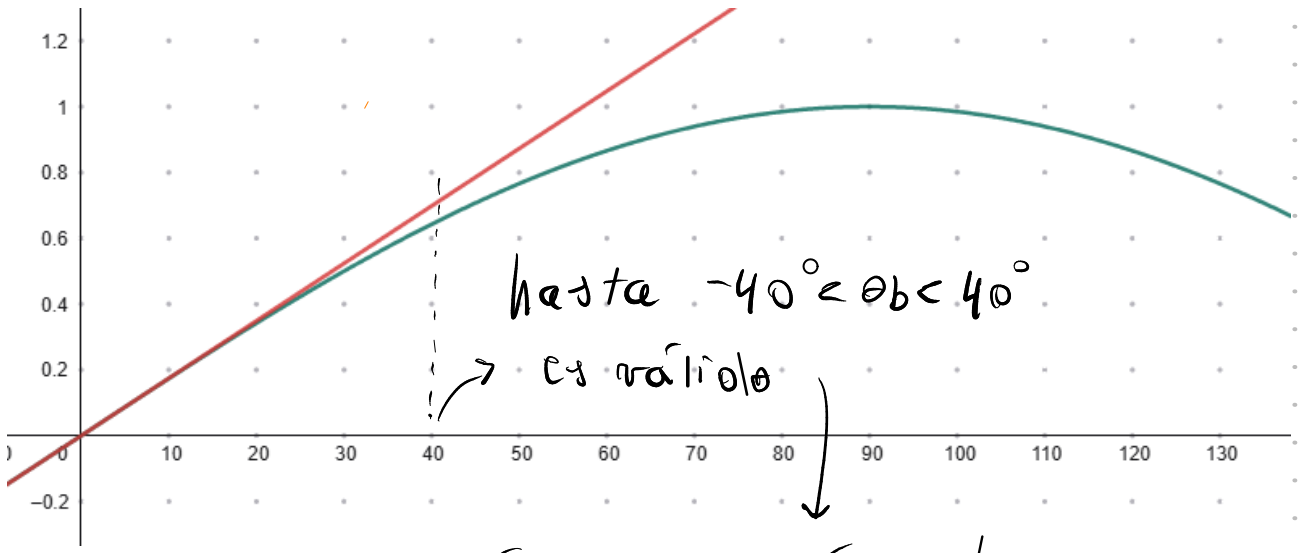
$N = mg \cdot \cos \theta_b$

$\theta_b \sim$ Chico $\rightarrow \operatorname{sen} \theta_b \cong \theta_b$

$\rightarrow \ddot{x} = g \cdot \theta_b$

$s^2 \cdot X(s) = g \cdot \Theta_b(s)$

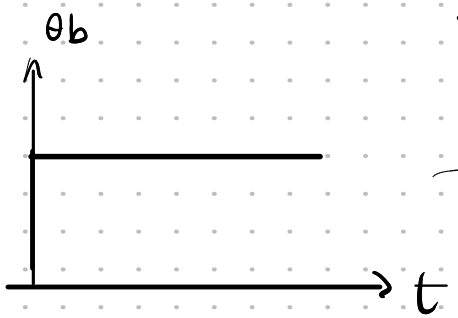
$X(s) / \Theta_b(s) = g / s^2$



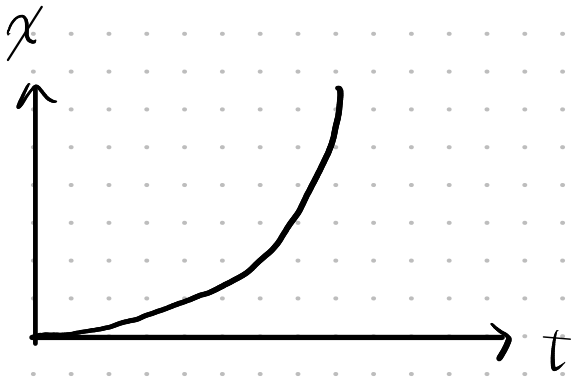
hasta $-40° < \theta_b < 40°$

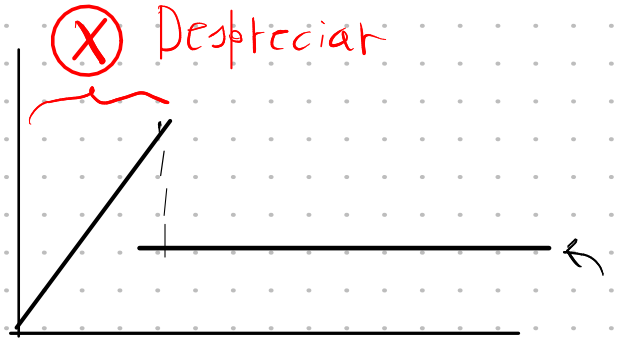
es válido

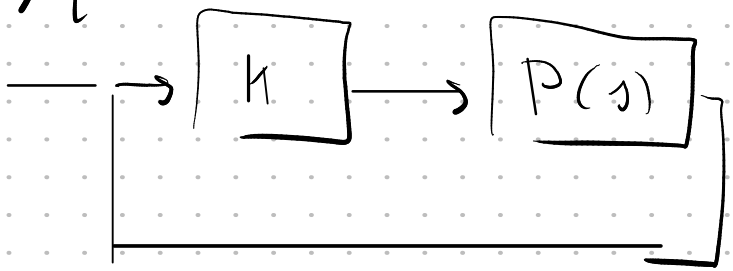
Seguro está en los límites de θ_b



X / Θ_b







Segundo modelo: $\mu_e = 0$; $\mu_d \neq 0$

$$m \cdot \ddot{x} = m \cdot g \cdot \operatorname{sen}\theta_b - \mu_d \cdot N$$

$$N = m \cdot g \cdot \cos\theta_b$$

$$\cancel{m} \cdot \ddot{x} = \cancel{m} g \operatorname{sen}\theta_b - \mu_d \cdot \cancel{m} \cdot g \cdot \cos\theta_b \leftarrow !$$

Este coseno nos mata

$$\begin{cases} \operatorname{sen}\theta_b \cong \theta_b \\ \cos(\theta_b) \cong 1 \end{cases}$$

$$s^2 X(s) = g \cdot \Theta_b - \mu_d \cdot g \frac{1}{s}$$

$$x, \dot{x}, \theta_b, \dot{\theta}_b$$

$$s^3 X(s) = g \cdot \Theta_b(s) \cdot s - \mu_d g$$

$$s^3 X(s) - g \Theta_b(s) \cdot s = -\mu_d \cdot g \quad ¿?$$

$$\begin{pmatrix} \dot{x} \\ \ddot{x} \end{pmatrix} = A \begin{pmatrix} x \\ \dot{x} \end{pmatrix} + B \cdot \theta_b$$

$$\left(s^3 \frac{X}{\Theta} - s \cdot g \right) \Theta = -\mu_d \cdot g$$

$$x = (1 \ 0) \begin{pmatrix} x \\ \dot{x} \end{pmatrix}$$

$$\ddot{x} = g \, \theta_b$$

$$\dot{x} = ¿?$$

Modelo 3

$$m \cdot \ddot{x} = m \cdot g \cdot \operatorname{sen} \theta_b - b \cdot \dot{x}$$

$$s^2 X(s) = g \, \Theta_b(s) - \frac{b}{m} \cdot s \cdot X(s)$$

$$(s + b/m) s \cdot X(s) = g \, \Theta_b(s)$$

$$X(s) / \Theta_b(s) = \frac{g}{(s + b/m) \cdot s}$$

$m = 33 \cdot 10^{-3}$ kg

$b/m = 0{,}91$

$b = 30{,}03 \cdot 10^{-3}$ kg/s

$\theta_b = 10^\circ \rightarrow P_x = 33g \cdot 10 \cdot \operatorname{sen} 10^\circ = 57{,}3 \cdot 10^{-3}$ N

Si el carrito se mueve a 10 cm/s (0,1 m/s)

la fuerza viscosa es $F_{vis} = 3 \cdot 10^{-3}$ N

¿Despreciable?